

Thevenin's Theorem

Paul John Placer

June 10, 2026

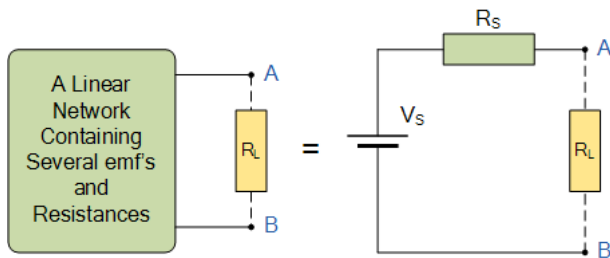
Thevenin's Theorem

Any *linear circuit* can be simplified into an *equivalent circuit* consisting of a single voltage source and series resistance connected to a load.

This simplification allows for easily analyzing the effect of the circuit on different loads without having to recalculate with the whole circuit for each load.

A *linear circuit* is a circuit consisting of only linear components: resistors, capacitors, and inductors; their voltage-current relationship is linear.

Introduction



Source: ilearnengineering.com

Thevenin Equivalent

The Thevenin equivalent circuit of any given circuit consists of the *Thevenin voltage* V_{Th} and the *Thevenin resistance* R_{Th} , all in series with the load component (usually modeled as a load resistor R_L).

To solve for V_{Th} and R_{Th} , first *disconnect the load*. These values can then be solved for by viewing the circuit through the resulting *open terminals* from the disconnected load.

Thevenin Equivalent

Thevenin Voltage

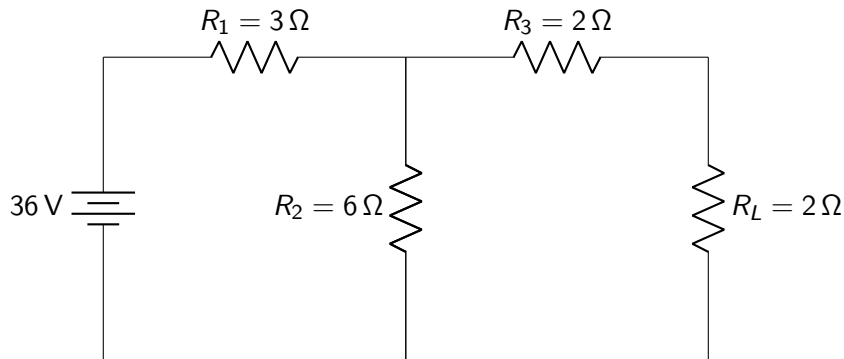
The Thevenin voltage V_{Th} is obtained by calculating the voltage between open terminals A and B .

Thevenin Resistance

The Thevenin resistance R_{Th} is obtained by zeroing all independent sources and then calculating the equivalent resistance of the resulting circuit as seen from open terminals A and B .

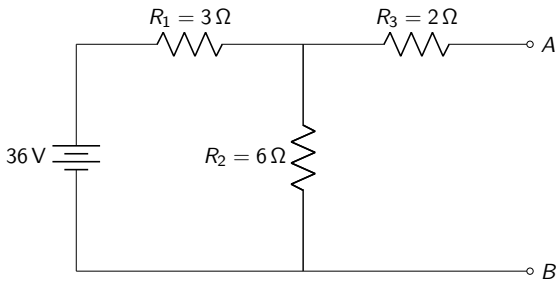
Example

Using Thevenin's Theorem, determine the voltage across R_L :



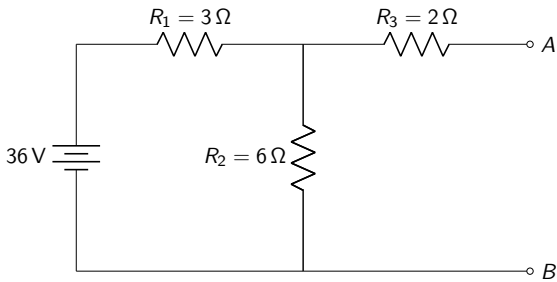
Example

Solving for V_{Th} :



Example

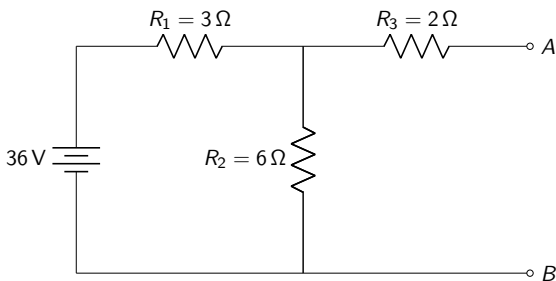
Solving for V_{Th} :



$$V_{Th} = V_{R_2}$$

Example

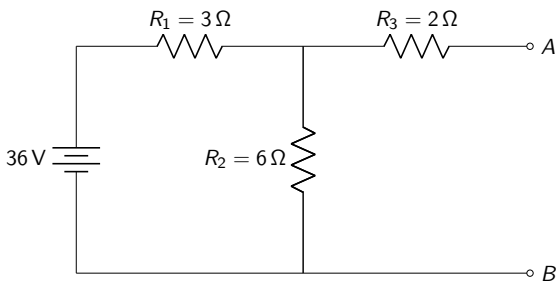
Solving for V_{Th} :



$$\begin{aligned} V_{Th} &= V_{R_2} \\ &= 36\text{ V} \frac{6\Omega}{3\Omega + 6\Omega} \end{aligned}$$

Example

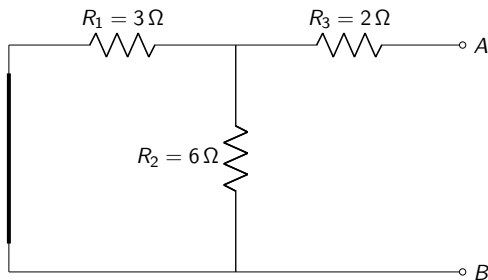
Solving for V_{Th} :



$$\begin{aligned} V_{Th} &= V_{R_2} \\ &= 36\text{ V} \frac{6\Omega}{3\Omega + 6\Omega} \\ &= 24\text{ V} \end{aligned}$$

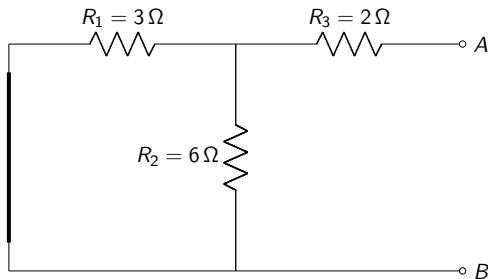
Example

Solving for R_{Th} :



Example

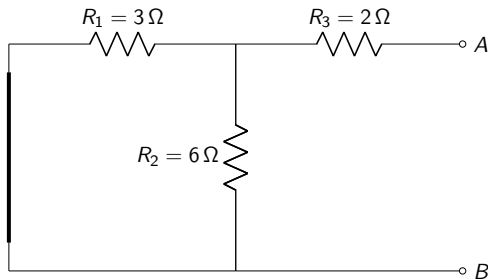
Solving for R_{Th} :



$$R_{Th} = 3\Omega \parallel 6\Omega + 2\Omega$$

Example

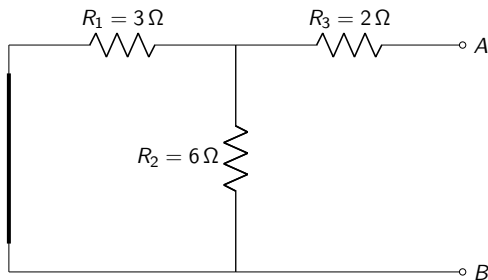
Solving for R_{Th} :



$$\begin{aligned} R_{Th} &= 3\Omega \parallel 6\Omega + 2\Omega \\ &= 2\Omega + 2\Omega \end{aligned}$$

Example

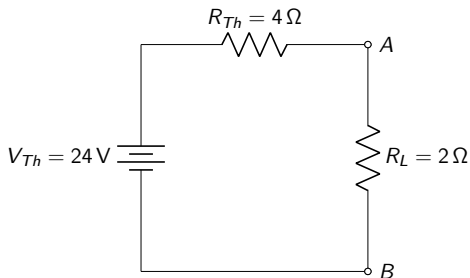
Solving for R_{Th} :



$$\begin{aligned} R_{Th} &= 3\Omega \parallel 6\Omega + 2\Omega \\ &= 2\Omega + 2\Omega \\ &= 4\Omega \end{aligned}$$

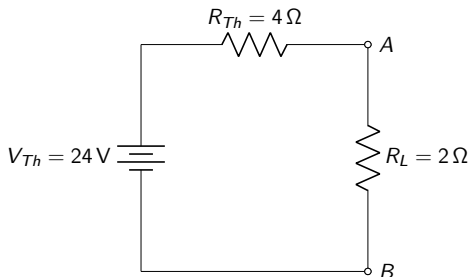
Example

The resulting Thevenin equivalent circuit:



Example

The resulting Thevenin equivalent circuit:

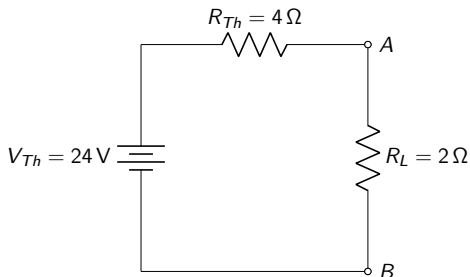


V_{R_L} becomes a simple *voltage divider*:

$$V_{R_L} = V_{Th} \frac{R_L}{R_{Th} + R_L}$$

Example

The resulting Thevenin equivalent circuit:

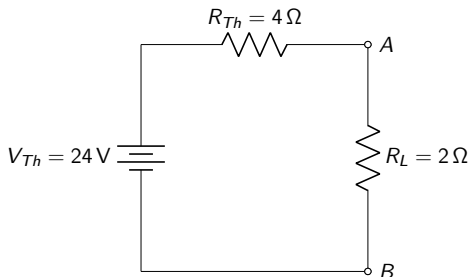


V_{R_L} becomes a simple *voltage divider*:

$$\begin{aligned} V_{R_L} &= V_{Th} \frac{R_L}{R_{Th} + R_L} \\ &= 24\text{ V} \frac{2\ \Omega}{4\ \Omega + 2\ \Omega} \end{aligned}$$

Example

The resulting Thevenin equivalent circuit:

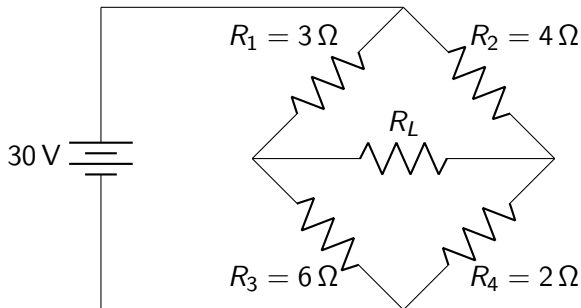


V_{R_L} becomes a simple *voltage divider*:

$$\begin{aligned} V_{R_L} &= V_{Th} \frac{R_L}{R_{Th} + R_L} \\ &= 24\text{ V} \frac{2\ \Omega}{4\ \Omega + 2\ \Omega} \\ &= 8\text{ V} \end{aligned}$$

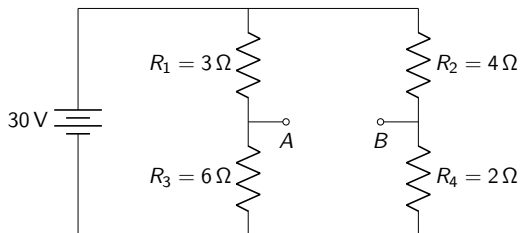
Example - Bridge Circuit

Find the Thevenin equivalent of the circuit below:



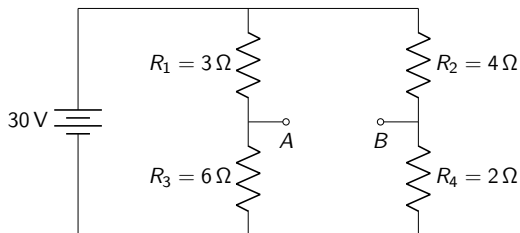
Example - Bridge Circuit

Solving for V_{Th} :



Example - Bridge Circuit

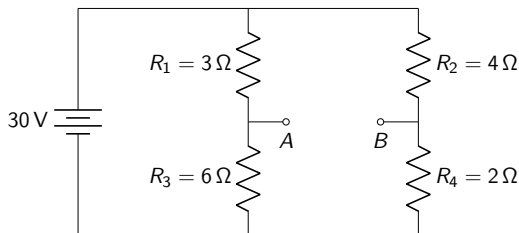
Solving for V_{Th} :



$$V_{Th} = V_A - V_B$$

Example - Bridge Circuit

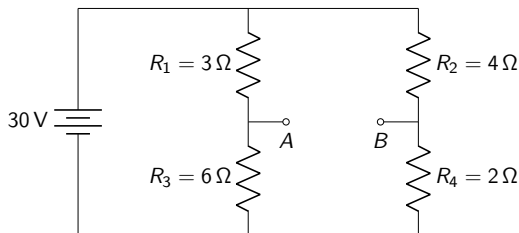
Solving for V_{Th} :



$$\begin{aligned} V_{Th} &= V_A - V_B \\ &= \frac{30\text{ V} \cdot 6\Omega}{3\Omega + 6\Omega} - \frac{30\text{ V} \cdot 2\Omega}{4\Omega + 2\Omega} \end{aligned}$$

Example - Bridge Circuit

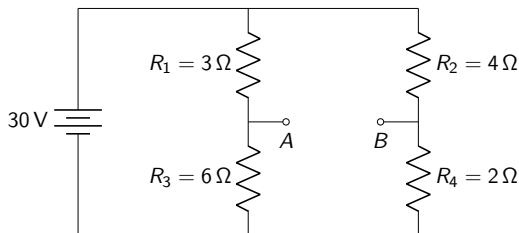
Solving for V_{Th} :



$$\begin{aligned} V_{Th} &= V_A - V_B \\ &= \frac{30\text{ V} \cdot 6\Omega}{3\Omega + 6\Omega} - \frac{30\text{ V} \cdot 2\Omega}{4\Omega + 2\Omega} \\ &= 20\text{ V} - 10\text{ V} \end{aligned}$$

Example - Bridge Circuit

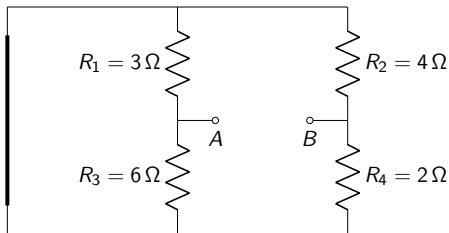
Solving for V_{Th} :



$$\begin{aligned} V_{Th} &= V_A - V_B \\ &= \frac{30\text{ V} \cdot 6\Omega}{3\Omega + 6\Omega} - \frac{30\text{ V} \cdot 2\Omega}{4\Omega + 2\Omega} \\ &= 20\text{ V} - 10\text{ V} \\ &= 10\text{ V} \end{aligned}$$

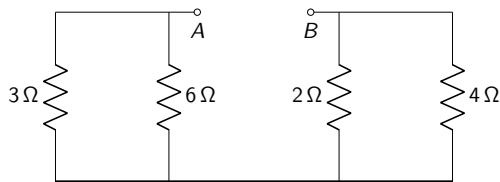
Example - Bridge Circuit

Solving for R_{Th} :



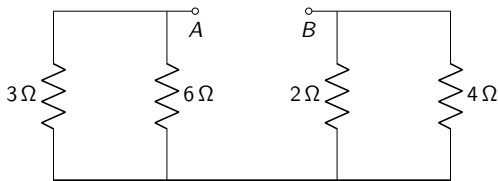
Example - Bridge Circuit

Solving for R_{Th} :



Example - Bridge Circuit

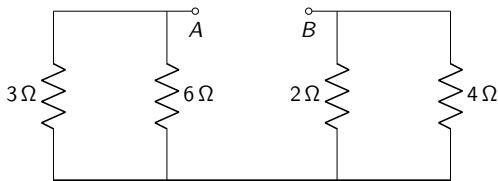
Solving for R_{Th} :



$$R_{Th} = 3\Omega \parallel 6\Omega + 4\Omega \parallel 2\Omega$$

Example - Bridge Circuit

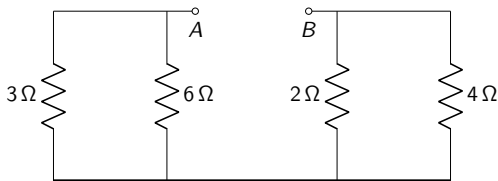
Solving for R_{Th} :



$$\begin{aligned} R_{Th} &= 3\Omega \parallel 6\Omega + 4\Omega \parallel 2\Omega \\ &= 2\Omega + 1.33\Omega \end{aligned}$$

Example - Bridge Circuit

Solving for R_{Th} :



$$\begin{aligned} R_{Th} &= 3\Omega \parallel 6\Omega + 4\Omega \parallel 2\Omega \\ &= 2\Omega + 1.33\Omega \\ &= 3.33\Omega \end{aligned}$$

Example - Bridge Circuit

The resulting Thevenin equivalent circuit:

